



REVIEW ARTICLE

# Continuous Kidney Replacement Therapy in Pediatric Intensive Care: A Practical Guide for Clinicians

Deblina Dasgupta<sup>1</sup> · Rajiv Sinha<sup>1</sup> · Akash Deep<sup>2</sup>

Received: 16 January 2025 / Accepted: 16 December 2025 / Published online: 3 February 2026  
© The Author(s), under exclusive licence to Indian Academy of Pediatrics 2026

## Abstract

As the complexities of children getting admitted to the pediatric intensive care unit (PICU) are increasing, so is the incidence of acute kidney injury (AKI) as well as the need for performing kidney replacement therapy (KRT). This necessitates special infrastructure as the standard extracorporeal therapy like haemodialysis is often not feasible in these children. Continuous kidney replacement therapy (CKRT) has become the KRT modality of choice in PICUs across high-income countries. Even in emerging economies like India, its use is increasing because of its advantages in critically sick children. This review article summarises the practical aspects of CKRT and serves as a concise guide for postgraduate trainees and intensivists working in the PICU.

**Keywords** Acute kidney injury · Critically ill children · Renal replacement therapy

## Introduction

The epidemiology of acute kidney injury (AKI) in children has evolved over recent years, with an increasing incidence reported from pediatric intensive care units (PICUs). Multicentre cohort studies have documented AKI in 26.9% of children admitted to PICUs and in 29.9% of neonates in neonatal intensive care units (NICUs) [1, 2]. Approximately 6% of critically ill children require kidney replacement therapy (KRT) [1]. Owing to the superior haemodynamic stability it offers, continuous kidney replacement therapy (CKRT) has emerged as the preferred modality in high-income countries and is being increasingly adopted in lower- and middle-income settings as well [3]. This article aims to equip postgraduate trainees rotating through the PICU with a practical working knowledge of CKRT.

## Principle

Continuous kidney replacement therapy is an extracorporeal modality that facilitates gradual removal of fluid and solutes over a prolonged duration (24 h/day). This gentle and continuous nature makes CKRT the preferred modality in children with haemodynamic instability and/or raised intracranial pressure [4].

Fluid and solute removal during CKRT occurs through the following mechanisms:

**Diffusion:** Movement of solutes across a semipermeable membrane from an area of higher to lower concentration. Solute clearance is enhanced by a higher concentration gradient, smaller molecular size of the solutes, and larger membrane pore size.

**Ultrafiltration:** Movement of solvent across a semipermeable membrane driven by a transmembrane pressure gradient.

**Convection:** Solutes are removed by solvent drag generated during ultrafiltration, allowing clearance of both small and middle molecular weight solutes. Increasing ultrafiltration enhances solvent drag; haemodynamic stability is maintained by the administration of replacement fluid.

**Adsorption:** Removal of solutes through chemical interactions with the membrane surface. For instance, the negatively charged AN69 membrane used in some CKRT filters

✉ Rajiv Sinha  
rajivsinha\_in@yahoo.com

<sup>1</sup> Division of Pediatric Nephrology, Institute of Child Health, Kolkata, West Bengal, India

<sup>2</sup> Department of Pediatric Critical Care, King's College Hospital, London, UK

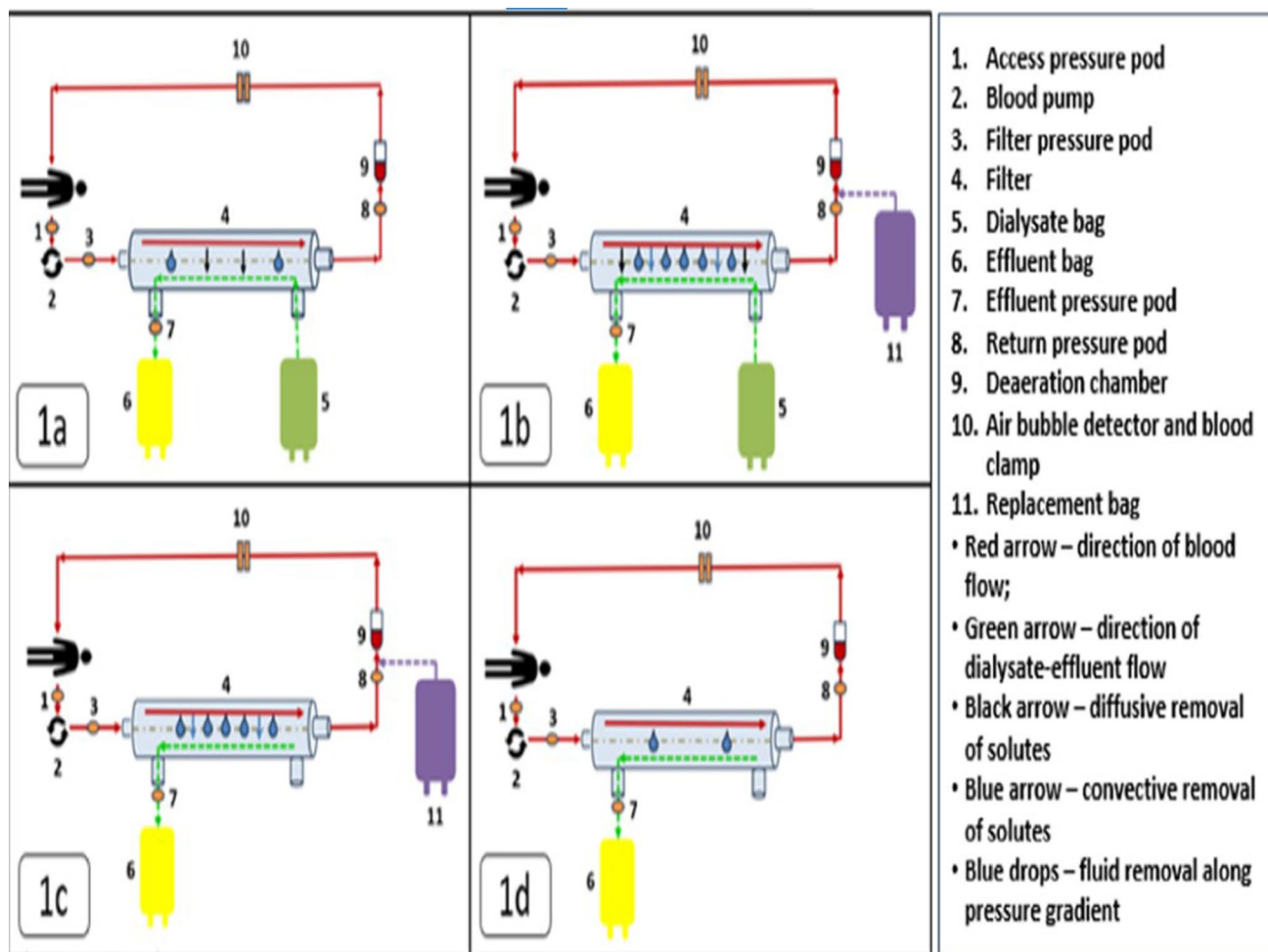
can bind positively charged molecules such as cytokines, thereby facilitating their clearance.

The four different modalities of CKRT and their primary mode of fluid/solute removal are elaborated in Fig. 1 and their advantages and disadvantages are summarised in Table 1.

## Indications

**Renal indications:** Standard indications for initiating CKRT in AKI include:

- Life threatening fluid overload ( $\geq 10\text{--}15\%$ ) with/without electrolyte/acid–base imbalance refractory to medical management: Fluid resuscitation in critically ill children results in fluid accumulation which can be assessed by the positive fluid balance developed since admission. When associated with peripheral or pulmonary oedema, it is usually referred to as fluid overload. The degree of fluid overload is objectively assessed by the fluid overload percentage, which is the cumulative positive balance expressed as percentage of the body weight, as depicted by the following formula:



**Fig. 1** Circuit diagram of various modes of Continuous Kidney Replacement Therapy (CKRT). **a** Continuous veno-venous hemodialysis (CVVHD) provides only diffusive removal through dialysis ± ultrafiltration. The counter current flow of dialysate from the dialysate bag (5) through the filter (4) helps achieve diffusive removal of solutes from blood to dialysate; removes predominantly small molecules as in hyperammonemia due to inborn error of metabolism. **b** CVVHDF combines diffusive removal through dialysis with convective removal through hemofiltration ± ultrafiltration. Here, high rates of ultrafiltration across the filter membrane (4) help achieve convective removal/hemofiltration; the lost fluid is replenished back into the circuit from the replacement bag (11) connected to the circuit

either pre-filter or post-filter (as in the diagram b), thereby helping to maintain haemodynamic stability; removes both small and middle molecules as in AKI, acute liver failure. **c** CVVH where only the replacement bag (11) is connected to the circuit to achieve convective removal of the solute through hemofiltration and the dialysate bag is absent; removes middle molecule effectively as in acute liver failure. **d** SCUF where only ultrafiltration removes excess fluid. Neither dialysis nor hemofiltration takes place; so dialysate bag and the replacement bag is absent. Useful in diuretic refractory fluid overload as in congestive cardiac failure and resistant oedema in nephrotic syndrome

**Table 1** Comparison of different modalities of KRT

| Modality                           | PD                          | HD                          | SLED                        | Continuous Kidney Replacement Therapy (CKRT) |                                        |                             |                 |
|------------------------------------|-----------------------------|-----------------------------|-----------------------------|----------------------------------------------|----------------------------------------|-----------------------------|-----------------|
|                                    |                             |                             |                             | CVVHD                                        | CVVHDF                                 | CVVH                        | SCUF            |
| Type                               | Paracorporeal               | Extracorporeal              | Extracorporeal              | Extracorporeal                               | Extracorporeal                         | Extracorporeal              | Extracorporeal  |
| Duration                           | 24 h                        | 4–6 h                       | 8–12 h                      | 24 h                                         | 24 h                                   | 24 h                        | 24 h            |
| Mechanism of clearance             | Diffusion, ultra-filtration | Diffusion, ultra-filtration | Diffusion, ultra-filtration | Diffusion, ultra-filtration                  | Convection, diffusion, ultrafiltration | Convection, ultrafiltration | Ultrafiltration |
| Control over fluid removal         | ±                           | +                           | ++                          | +++                                          | +++                                    | +++                         | +++             |
| Rapidity of small molecule removal | +                           | +++                         | ++                          | ++                                           | ++                                     | ++                          | Not applicable  |
| Middle molecule removal            | –                           | –                           | –                           | –                                            | +                                      | ++                          | Nil             |
| Haemodynamic tolerance             | +++                         | –                           | ++                          | +++                                          | +++                                    | +++                         | +++             |
| Technical challenges               | –                           | +                           | ++                          | +++                                          | +++                                    | +++                         | +++             |
| Need for anticoagulation           | –                           | +                           | ++                          | +++                                          | +++                                    | +++                         | +++             |
| Ease in neonates and small infants | ++                          | –                           | –                           | +                                            | +                                      | +                           | +               |

CVVHD continuous veno-venous hemodialysis, CVVHDF continuous veno-venous hemodiafiltration, CVVH continuous veno-venous hemofiltration, HD hemodialysis, PD peritoneal dialysis, SCUF slow continuous ultra filtration, SLED sustained low efficiency dialysis

$$\text{Fluid overload (\%)} = \frac{\text{Total fluid in (L)} - \text{Total fluid out (L)}}{\text{Body weight (Kg)}} \times 100$$

In oligo-anuric children, one might have to initiate KRT at lesser degree of fluid overload to create space for necessary fluid and nutritional support.

- Uremic encephalopathy
- Uremic pericarditis

*Non-renal indications:* These are usually in situations where a sustained prolonged removal of toxin/toxic metabolites is required, and includes:

- Acute neurological crisis in inborn errors of metabolism (IEMs) for the removal of toxic metabolites like ammonia, organic acids
- *Acute liver failure:* In the presence of hyperammonemia ( $> 150 \mu\text{mol/L}$  or  $> 256 \mu\text{g/dL}$ ), or stage 3 or 4 hepatic encephalopathy as a part of bridge therapy to liver transplantation
- *Sepsis associated with multi-organ dysfunction syndrome (MODs):* Removal of cytokines/endotoxins with filters like Oxiris
- Tumour lysis syndrome and rhabdomyolysis

- *Poisoning:* Removal of dialyzable drugs like barbiturates, salicylates, metformin, lithium, valproate, etc., and toxins like glyphosate

In these conditions, CKRT is preferred as it prevents the rebound increase in the levels of toxic solutes like ammonia, a complication often seen with other extracorporeal modalities of shorter duration like hemodialysis (HD) and sustained low efficiency dialysis (SLED). More accurate control over the clearance of these substances makes it more desirable than peritoneal dialysis, especially in presence of haemodynamic instability and/or raised intracranial pressure, which are often found to complicate the clinical scenario in such cases. Immunomodulatory effect by sustained removal of pro-inflammatory cytokines is another advantage CKRT provides in children with acute liver failure and sepsis.

### Timing

The definitions of ‘early’ or ‘late’ initiation used in various studies are unclear and variable [5–7]. Retrospective studies have shown that early initiation of KRT in children with AKI may decrease mortality [8, 9]. However, early



initiation carries the risk of over-treatment, exposing the child to potential complications of CKRT and increased financial burden. Current evidence points towards fluid overload (> 10%) being one of the most important factors with evidence suggesting an increase in mortality by 3% for every 1% increase in fluid overload at the time of initiation of CKRT [10].

## Logistics

Apart from trained intensivists, critical care nephrology specialists, and proficient nursing team, important components essential for CKRT include:

**Vascular access:** A good vascular access is key to ensuring optimum CKRT functioning. Appropriate size (Table 2) non-tunnelled, non-cuffed, double lumen haemodialysis catheters are inserted under guidance using ultrasound (USG) [4]. The catheters should be wide enough to provide low resistance, but at the same time the catheter/vessel diameter ratio should not exceed 0.45 to prevent central vein thrombosis. The catheter should be long enough so that the tip resides in the cavo-atrial junction for catheters placed in internal jugular vein (IJV), or in the inferior vena cava for catheters placed in the femoral veins, as confirmed by a radiograph before initiating CKRT [4]. The right IJV is the preferred site of insertion due its straight course, followed

by the femoral veins specially in presence of deranged coagulation parameters [4, 11]. The left IJV is not preferred because of its angulation, while the subclavian veins should be avoided as it is associated with an increased incidence of complications like stenosis of the vein which may prevent subsequent placement of vascular access for long-term maintenance haemodialysis.

**CKRT machine:** Currently in India, machines manufactured by Baxter (Prismaflex, Prisma) and Fresenius (Fresenius Multifiltrate, Fresenius Multifiltrate AV Pro) are more easily available. Newer machines with lower extracorporeal volumes, such as CARPEDIEM (Cardio-Renal Pediatric Dialysis Emergency Machine), NIDUS (Newcastle Infant Dialysis and Ultrafiltration System), and AQUADEX are still not available in India.

**Integrated hemofilter and tubing set:** Similar to conventional HD filters, CKRT filters have a hollow fibre design but with larger pores allowing removal of substances with larger MW (30–50 KDa vs up to 15 KDa in high-flux HD). They are made up of biocompatible synthetic materials like polysulfone, AN-69, etc. A range of filters and tubing sets should be available to accommodate children of all age groups as shown in Table 3. The circuit volume should not exceed 10% of the child's circulating blood volume (Total blood volume = 80–90 mL/kg in full-term neonates, 90–100 mL/kg in preterm neonates, 70–80 mL/

**Table 2** Vascular catheter sizes as per the weight of the child. Adapted from Bunchman et al. [11]

| Weight range | Vascular catheter size                                                                                                                          | Preferred site                              |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Neonate      | Single Lumen 5 F (Cook Medical, Bloomington, IN), Double Lumen 6.5 F (Arrow, Reading, PA),<br>Double lumen 7 F (Cook Medcomp, Harleysville, PA) | Femoral vein                                |
| 3–6 kg       | Double Lumen 7 F (Cook/Medcomp), Double Lumen 6.5 F (Arrow), Triple Lumen 7 F (Medcomp/Arrow)                                                   | Right IJV > Femoral > left IJV > Subclavian |
| 6–30 kg      | Double lumen 8 F (Kendall, Mansfield, MA; Arrow)                                                                                                | Right IJV > Femoral > left IJV > Subclavian |
| > 30 kg      | Double lumen 10 F (Arrow, Kendall) or Triple lumen 12 F (Arrow, Kendall)                                                                        | Right IJV > Femoral > left IJV > Subclavian |

IJV Internal Jugular Vein

Usually 5 F catheter is not preferred with the smallest catheter preferred is 6.5 F because of longer catheter life

**Table 3** Different filter sizes available in India with corresponding priming volumes

| Brand                       | Name    | Surface area        | Priming volume of set/filter | Minimum blood flow rate (ml/min) | Minimum patient weight (kg)             |
|-----------------------------|---------|---------------------|------------------------------|----------------------------------|-----------------------------------------|
| Vantive (previously Baxter) | HF20    | 0.2 m <sup>2</sup>  | 58 ml (set)                  | 20                               | 8                                       |
|                             | M60     | 0.6 m <sup>2</sup>  | 93 ml (set)                  | 50                               | 11                                      |
|                             | M100    | 0.9 m <sup>2</sup>  | 152 ml (set)                 | 75                               | 30                                      |
|                             | M150    | 1.5 m <sup>2</sup>  | 189 ml (set)                 | 100                              | 30                                      |
| Fresenius                   | AVPaed  | 0.2 m <sup>2</sup>  | 18 ml (filter)               | 10                               | Not mentioned in product specifications |
|                             | AV400S  | 0.75 m <sup>2</sup> | 52 ml (filter)               | 50                               |                                         |
|                             | AV600S  | 1.4 m <sup>2</sup>  | 100 ml (filter)              | 100                              |                                         |
|                             | AV1000S | 1.8 m <sup>2</sup>  | 130 ml (filter)              | 200                              |                                         |



kg in infants, 70–75 mL/kg in children, and 65–70 mL/kg in adults) and the surface area of the filter should be 0.8–1 times the child's body surface area, i.e. BSA).

**Circuit priming:** Priming of the circuit is indicated in haemodynamic instability, anaemia or weight < 10 kg. Priming solutions include crystalloids, 5% albumin or blood. Blood priming can cause metabolic complications (acidosis, hyperkalaemia, hypocalcaemia), especially in infants < 5 kg, and can precipitate bradykinin release syndrome with AN-69 membranes. Ways to circumvent these problems include: (a) using blood buffered with sodium bicarbonate, (b) pre-dialysing the blood primed circuit before connecting to the patient, or (c) priming the circuit with crystalloid or 5% albumin, and transfusing blood to the child through an intravenous cannula.

**Dialysis and replacement solutions:** The dialysate and replacement solutions can be used interchangeably and contain:

- (a) **Electrolytes:** sodium (130–140 meq/L), potassium (0–4 meq/L), chloride (108–120 meq/L), magnesium (1–1.5 meq/L), and calcium (0–3.5 meq/L)
- (b) **Buffer:** lactate (3 meq/L) and bicarbonate (22–32 meq/L) in lactate-based solutions or only bicarbonate (25–35 meq/L) in bicarbonate-based solutions
- (c) **Dextrose** (0–1.1 g/dL)

The compositions of the fluids are different if regional citrate anticoagulation is used. Bicarbonate-based solutions are preferred in patients with circulatory shock, liver failure, or lactic acidosis [4].

**Anticoagulation:** Multiple factors play important roles in circuit clotting including vascular access, dose of pre-dilution used, usage of blood products, and patient-related factors. Anticoagulation prolongs circuit life, decreases treatment interruptions and downtimes, prevents blood loss, and decreases financial burden. The two most commonly used anticoagulation methods are systemic anticoagulation with unfractionated heparin (UFH) and regional citrate anticoagulation (RCA). The other less commonly used ones are detailed in Web Table 1. RCA is the preferred anticoagulation for CKRT unless there are contraindications for the same such as severe liver impairment or shock with muscle hypoperfusion [4]. In view of the steep learning curve and labour-intensive monitoring with RCA, UFH is still the most common anticoagulant used in India. Sometimes in patients with grossly deranged coagulation and non-availability of alternatives like RCA or prostacyclin, heparin-free CKRT needs to be done with regular saline flushes of the circuit. In these situations, circuit life can be prolonged by the following steps:

- (a) Optimising vascular access (appropriate size/avoiding kinks or angulations), and promptly responding to alarms
- (b) Optimising filter size
- (c) Adjustments in CKRT prescription like minimising convection and preferentially using dialysis when the condition permits, using prefilter dilution, keeping filtration fraction < 20–25%, and minimising air–blood contact in the deaeration chamber
- (d) Normal saline flushes equivalent to the circuit volume, administered every 30–60 min, and ensuring that these saline flushes are accounted for in the ultrafiltration calculation

**Blood warmer:** Newer machines have integrated blood warmers in them. If not present, blood warmers should be attached separately to the circuit as hypothermia can be a serious complication, especially in neonates and infants.

## Prescription

### Dose

The dose in CKRT equals to the effluent flow rate which is the sum of dialysate flow rate (diffusive clearance), if used, and the replacement flow rate (convective clearance). Robust data regarding the optimal CKRT dose in pediatric population is lacking. Kidney Disease: Improving Global Outcomes (KDIGO) AKI guidelines 2012 recommend a combined delivered CKRT dose of 20–25 mL/kg/hour in AKI as large multicentre RCTs in adults have failed to show benefits of higher doses [4, 12, 13]. Other recommendations have suggested basing the dose on BSA and targeting a minimum of 2000 mL/1.73m<sup>2</sup>/hour [11]. As the final delivered dose is usually less than the prescribed dose due to CKRT downtime (secondary to factors such as circuit changes, bag changes, and time lost addressing alarms, vascular access issues, and temporary discontinuation of CKRT for off-room medical procedures), it is prudent to start with higher dose of 25–30 mL/kg/hour, and keep the delivered dose-to-prescribed dose ratio  $\geq 0.8$  [4]. In certain scenarios, such as hyperammonemia due to IEMs, acute liver failure, hyperkalaemia in tumour lysis syndrome, and certain intoxications, higher doses of up to 8000 mL/1.73m<sup>2</sup>/hour have been used.

### Prescription Parameters

**Mode** (Fig. 1): Include Continuous Veno-Venous Hemodialysis (CVVHD), Continuous Veno-venous Hemofiltration (CVVH), Continuous Veno-venous Hemodiafiltration (CVVHDF), and Slow Continuous Ultrafiltration (SCUF).





Selection of mode depends on the indication, availability, and centre preference.

**Circuit size:** Depends on the child's BSA and blood volume (Table 3).

**Blood flow rate:** Usually initiated at 3–5 mL/kg/min and then increased to 8–10 mL/kg/min if haemodynamically tolerated. Higher blood flow improves clearance of solutes and decreases chances of filter clotting.

**Dialysate flow rate:** It is usually kept at 50% of the prescribed dose in CVVHDF and 100% of the prescribed dose in CVVHD (vide case scenarios).

**Replacement flow rate:** It is 50% of the prescribed dose in CVVHDF and 100% of the prescribed dose in CVVH (vide case scenarios). This replacement fluid is further divided into prefilter and postfilter fractions (Fig. 1). The prefilter fraction helps dilute the blood going into the filter, reducing the chances of filter clotting, but also decreases the efficacy of diffusive clearance. By convention, some centres prefer a prefilter-to-postfilter fraction ratio of 50:50, while others prefer a ratio of 70:30. The postfilter fraction cannot be zero, as this can cause mixing of air with blood in the deaeration chamber, and clotting in return line.

**Fluid removal:** In a haemodynamically unstable child, patient fluid removal is initially kept at 0 mL/hour and then gradually increased further, if tolerated, aiming for a neutral balance. As CKRT progresses, improvement in the acidosis is anticipated which should lead to better haemodynamic stability, thereby allowing gradual removal of accumulated fluid over next few days.

**Anticoagulation:** Based on the patient's clinical condition and the centre's experience and protocol (Web Table 1).

Filtration fraction (usually displayed on the monitor) should be  $\leq 20$ –25% to minimise chances of circuit clotting. Measures that can help to achieve this includes: increasing blood flow rate, dialysate flow rate, and the prefilter fraction of the replacement flow rate along with decreasing total replacement flow rate. These are especially important where no anticoagulation is being used. Web Table 2 gives example of standard CKRT prescriptions in some common scenarios.

## Monitoring

Monitoring of clinical, laboratory, and technical parameters is essential as CKRT can potentially cause life-threatening complications (Web Tables 3 and 4).

## Complications

- Fluid and electrolyte imbalance
- Hypothermia
- Complications related to anticoagulation: bleeding, heparin-induced thrombocytopenia, hypersensitivity reaction, metabolic complications with RCA

- Circuit clotting and blood loss
- Vascular access-related complications: catheter related bloodstream infection, hemothorax, pneumothorax, aneurysms, etc.
- Thromboembolism
- Membrane reactions—Bradykinin release syndrome can happen with the use of AN-69 membranes especially with blood priming/concomitant use of ACE inhibitors. Hypersensitivity reactions can happen with all types of membranes.
- Loss of nutrients, and drugs in the effluent

## Nutrition

The advantage of creating fluid spaces by removal of water by CKRT should be utilised by optimising nutrition. One should account for ICU status such as reducing calorie target by 5–10% if ventilated as well as accounting for calories in dialysate/replacement fluid. Due to protein catabolism and loss of amino acids in CKRT effluent, an augmented protein intake has been recommended: 0–2 years old—3–3.5 g/kg/day, 2–13 years—2–2.5 g/kg/day and 13–18 years—2 g/kg/day.

## Drug Dosing Considerations

In children receiving CKRT, the pharmacokinetics of drugs are hugely altered, partly due to alterations in patient characteristics (altered kidney and liver functions, changes in body water composition, changes in plasma protein binding) and mostly due to extracorporeal clearance of the drugs, which in turn is governed by the drugs' physicochemical properties (influencing their volume of distribution, plasma protein binding, water solubility) and CKRT specifications (modality, membrane characteristics, and flow rates). These pharmacokinetic alterations need to be considered for guiding drug dosing during CKRT, to avoid the hazards of underdosing or overdosing, especially for antimicrobials, antiepileptic drugs (AEDs), sedatives, analgesics and antiarrhythmics. For example, drugs like meropenem, vancomycin, and piperacillin have significant renal clearance and also are freely filtered during CKRT. As a result, their clearance in children on CKRT is more than their clearance in children with AKI not on CKRT; hence dosing of these antimicrobials will have to be higher in children on CKRT than in the latter group to attain adequate therapeutic concentrations, more so when CKRT is being done due to non-renal indications in the absence of AKI [14]. Amongst AEDs, those with lesser degree of plasma protein binding like levetiracetam are significantly cleared by CKRT. The pharmacokinetics of highly plasma protein bound AEDs like phenytoin, valproate also get significantly altered due to high urea levels during AKI competing for the plasma proteins, thereby increasing



the free fractions of these drugs in blood and predisposing to their adverse effects [15]. Presently, there is lack of robust evidence backing the dosing recommendations for many drugs in children with CKRT. Ideally, the dosing should be guided by therapeutic drug monitoring whenever possible. Dosing of commonly used antimicrobials and antiepileptics is shown in Web Table 5.

### Discontinuation of CKRT

Similar to initiation, discontinuation also lacks robust evidence with spontaneous improvement in urine output  $> 0.5$  mL/kg/hour for 6–24 h correlating most strongly with successful discontinuation [16, 17]. KDIGO advises against the use of diuretics to promote kidney function recovery or decrease the duration of CKRT [8]. Often IHD/SLED is required as a bridge before full discontinuation [18].

### Challenges in Neonates and Small Infants

In neonates and small infants requiring KRT, CKRT is the preferred modality if the baby is having abdominal pathology such as intestinal perforation/necrotising enterocolitis (NEC) or recent open abdominal surgery. It is also the optimum mode when higher clearance rates are needed as in IEM. Performing CKRT in this cohort is technically challenging, owing to the difficulty in securing vascular access that allows optimum blood flow rates, and optimising extra corporeal blood volume [19]. US FDA does not approve the use of most of currently available CKRT machines in children weighing  $< 8$  kg. Newer machines with lower extracorporeal volumes and greater accuracy have been developed but are not widely available.

### Challenges in Resource-Limited Settings

Although CKRT has found widespread use across critical care units in developed nations, its use in LMICs like India remains restricted by factors like environmental factors, socioeconomic limitations, infrastructural constraints, lack of reimbursements for costly consumables, non-availability of trained health care professionals round the clock, lack of adequate technical and laboratory support, and administrative hindrances [20, 21]. CKRT machines cost around 24–30 lakh INR (27,000–30,000 USD). The initiation cost for CKRT (including cost of hemodialysis catheters, CKRT filter) can range from 24,000–34,000 INR (275–390 USD) approximately, with the cost of consumables needed thereafter (periodic changing of dialysis and replacement bags, etc.) ranging from 6500 to 30,000 INR (75–345 USD) approximately on a daily basis depending on the child's body weight. The costs for running CKRT with RCA are even

higher. To add to this, are the costs of regular monitoring of lab parameters. The financial constraints in resource-limited settings may be overcome with coverage under Government Health Schemes and/or funding from Non-Governmental Organisations (NGOs). Formulation of clinical practice guidelines and policies, earmarking specialised team for CKRT who are willing to shoulder the added responsibility, education and training of health care professionals with periodic reinforcements, ensuring in-house availability of technical support, multidisciplinary liaisons, periodic audit of performance data and outcome measures, and implementation of quality control measures are key to successful and sustainable adoption of CKRT programmes, even in resource-limited scenarios [20, 21].

### Outcome of CKRT

In hospital mortality rates of 35–64% have been reported from developed countries [3, 9, 10, 22–24], whereas the same reported from a tertiary care centre in India was 75% [25]. Associated comorbidity, underlying disease, severity of illness, associated MODS and degree of fluid overload at CKRT initiation are the main factors influencing mortality. Children requiring CKRT have also been found to require longer duration of mechanical ventilation and PICU stay [9, 22]. Children receiving CKRT due to severe AKI are also at increased risk of developing long-term adverse kidney consequences. Hence it is imperative to follow up these children long term for early detection of CKD or risk factors for CKD, although guidelines regarding the frequency or duration of the same are currently lacking.

### Advances Over the Last Decades

- (a) Membranes with adsorptive surfaces (Oxiris®) aid in pro-inflammatory states like sepsis with MODS by removing cytokines and endotoxins but conclusive evidence supporting mortality benefit is lacking [26, 27].
- (b) Selective cytopheretic device bind activated leucocytes and may show mortality benefit and reduce KRT dependency when used in critically ill patients with AKI in presence of hypocalcemia within circuit [28]. Safety and feasibility in pediatric patients have been demonstrated [29].
- (c) Extracorporeal CO<sub>2</sub> removal device helps promote lung protective ventilation strategies in children with hypercapnic respiratory failure [30].
- (d) CARPEDIEM has been approved in 2020 by the US FDA for use in children weighing 2.5–10 kg. Others like NIDUS and AQUADEX are in pipeline. These machines allow CKRT in neonates and small infants with very low blood flow rates (2–50 mL/min) through small calibre vascular accesses (4–5 Fr) [31–33].



- (e) CKRT filters with high cut-off (HCO) membranes allow removal of substances with molecular weights similar to albumin. Although conclusive proof is lacking, they may be beneficial for removal of cytokines in sepsis, cytokine storms, etc. [34].

## Conclusion

Although costly and labour-intensive, CKRT gives an edge in the management of critically ill children needing KRT. Its increasing adoption in pediatric intensive care, even in LMICs, requires standardisation of protocols and training to optimise outcomes and reduce complications.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s13312-026-00281-z>.

**Author Contributions** RS contributed to the study conception and design. All authors contributed to the writing of the article. All authors read and approved the final manuscript.

**Funding** None.

**Conflict of interest** None.

## References

- Kaddourah A, Basu RK, Bagshaw SM, Goldstein SL; AWARE Investigators. Epidemiology of acute kidney injury in critically ill children and young adults. *N Engl J Med*. 2017;376:11–20.
- Jetton JG, Boohaker LJ, Sethi SK et al. Neonatal Kidney Collaborative (NKC). Incidence and outcomes of neonatal acute kidney injury (AWAKEN): a multicentre, multinational, observational cohort study. *Lancet Child Adolesc Health*. 2017;1:184–94.
- Symons JM, Chua AN, Somers MJ, et al. Demographic characteristics of pediatric continuous renal replacement therapy: a report of the prospective pediatric continuous renal replacement therapy registry. *Clin J Am Soc Nephrol*. 2007;2:732–8.
- Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. *Kidney Inter Suppl*. 2012;2:1–138.
- Zarbock A, Kellum JA, Schmidt C, et al. Effect of early vs delayed initiation of renal replacement therapy on mortality in critically ill patients with acute kidney injury: the ELAIN randomized clinical trial. *JAMA*. 2016;315:2190–9.
- STARTR-AKI Investigators; Canadian Critical Care Trials Group; Australian and New Zealand Intensive Care Society Clinical Trials Group; United Kingdom Critical Care Research Group; Canadian Nephrology Trials Network; Irish Critical Care Trials Group; Bagshaw SM, Wald R, Adhikari NKJ, et al. Timing of initiation of renal-replacement therapy in acute kidney injury. *N Engl J Med*. 2020;383:240–51.
- Gaudry S, Hajage D, Martin-Lefevre L, et al. Comparison of two delayed strategies for renal replacement therapy initiation for severe acute kidney injury (AKIKI 2): a multicentre, open-label, randomised, controlled trial. *Lancet*. 2021;397:1293–300.
- Modem V, Thompson M, Gollhofer D, Dhar AV, Quigley R. Timing of continuous renal replacement therapy and mortality in critically ill children\*. *Crit Care Med*. 2014;42:943–53.
- Cortina G, McRae R, Hoq M, et al. Mortality of critically ill children requiring continuous renal replacement therapy: effect of fluid overload, underlying disease, and timing of initiation. *Pediatr Crit Care Med*. 2019;20:314–22.
- Sutherland SM, Zappitelli M, Alexander SR, et al. Fluid overload and mortality in children receiving continuous renal replacement therapy: the prospective pediatric continuous renal replacement therapy registry. *Am J Kidney Dis*. 2010;55:316–25.
- Bunchman TE, Brophy PD, Goldstein SL. Technical considerations for renal replacement therapy in children. *Semin Nephrol*. 2008;28:488–92.
- Bellomo R, Cass A, Cole L, et al. Intensity of continuous renal-replacement therapy in critically ill patients. *N Engl J Med*. 2009;361:1627–38.
- Palevsky PM, Zhang JH, O'Connor TZ, et al. Intensity of renal support in critically ill patients with acute kidney injury. *N Engl J Med*. 2008;359:7–20.
- Dubinsky S, Watt K, Saleeb S, et al. Pharmacokinetics of Commonly used medications in children receiving continuous renal replacement therapy: a systematic review of current literature. *Clin Pharmacokinet*. 2022;61:189–229.
- Mahmoud SH, Zhou XY, Ahmed SN. Managing the patient with epilepsy and renal impairment. *Seizure*. 2020;76:143–52.
- Katulka RJ, Al Saadon A, Sebastiani M, et al. Determining the optimal time for liberation from renal replacement therapy in critically ill patients: a systematic review and meta-analysis (DOnE RRT). *Crit Care*. 2020;24:50.
- Wei EY, Vuong KT, Lee E, Liu L, Ingulli E, Coufal NG. Predictors of successful discontinuation of continuous kidney replacement therapy in a pediatric cohort. *Pediatr Nephrol*. 2023;38:2221–31.
- Sinha R, Sethi SK, Bunchman T, Lobo V, Raina R. Prolonged intermittent renal replacement therapy in children. *Pediatr Nephrol*. 2018;33:1283–96.
- Askenazi DJ, Goldstein SL, Koralkar R, et al. Continuous renal replacement therapy for children  $\leq 10$  kg: a report from the prospective pediatric continuous renal replacement therapy registry. *J Pediatr*. 2013;162:587–92.e3.
- Srisawat N, Chakravarthi R. CRRT in developing world. *Semin Dial*. 2021;34:567–75.
- Sekhar JC, Nallasamy K, Jayashree M, et al. Continuous renal replacement therapy (CRRT) program initiation in PICU of a resource limited setting: a retrospective analysis of challenges and outcomes. *BMC Nephrol*. 2025;26:386.
- Hayes LW, Oster RA, Tofil NM, Tolwani AJ. Outcomes of critically ill children requiring continuous renal replacement therapy. *J Crit Care*. 2009;24:394–400.
- Stanski NL, Gist KM, Hasson D, et al. Characteristics and outcomes of children and young adults with sepsis requiring continuous renal replacement therapy: a comparative analysis from the Worldwide Exploration of Renal Replacement Outcomes Collaborative in Kidney Disease (WE-ROCK). *Crit Care Med*. 2024;52:1686–99.
- Choi SJ, Ha EJ, Jhang WK, Park SJ. Factors associated with mortality in continuous renal replacement therapy for pediatric patients with acute kidney injury. *Pediatr Crit Care Med*. 2017;18:e56–61.
- Sravani M, Krishnasamy S, Deepthi B, et al. Experience of continuous kidney replacement therapy in a tertiary care unit of a lower-middle-income country. *Pediatr Nephrol*. 2025;40:2083–90.
- Davies B, Cohen J. Endotoxin removal devices for the treatment of sepsis and septic shock. *Lancet Infect Dis*. 2011;11:65–71.
- Becker S, Lang H, Vollmer Barbosa C, Tian Z, Melk A, Schmidt BMW. Efficacy of CytoSorb®: a systematic review and meta-analysis. *Crit Care*. 2023;27:215.
- Tumlin JA, Galphin CM, Tolwani AJ, et al. SCD Investigator Group. A multi-center, randomized, controlled, pivotal study to





- assess the safety and efficacy of a selective cytopheretic device in patients with acute kidney injury. *PLoS One*. 2015;10:e0132482
29. Goldstein SL, Askenazi DJ, Basu RK, et al. Use of the selective cytopheretic device in critically ill children. *Kidney Int Rep*. 2020;6:775–84.
  30. Combes A, Fanelli V, Pham T, Ranieri VM. Feasibility and safety of extracorporeal CO<sub>2</sub> removal to enhance protective ventilation in acute respiratory distress syndrome: the SUPERNOVA study. *Intensive Care Med*. 2019;45:592–600.
  31. Ronco C, Garzotto F, Brendolan A, et al. Continuous renal replacement therapy in neonates and small infants: development and first-in-human use of a miniaturised machine (CARPEDIEM). *Lancet*. 2014;383:1807–13.
  32. Coulthard MG, Crosier J, Griffiths C, et al. Haemodialysing babies weighing <8 kg with the Newcastle Infant Dialysis and Ultrafiltration System (NIDUS): comparison with peritoneal and conventional haemodialysis. *Pediatr Nephrol*. 2014;29:1873–81.
  33. Askenazi D, Ingram D, White S, et al. Smaller circuits for smaller patients: improving renal support therapy with Aquadex. *Pediatr Nephrol*. 2016;31:853–60.
  34. Zhou Z, Kuang H, Wang F, Liu L, Zhang L, Fu P. High cut-off membranes in patients requiring renal replacement therapy: a systematic review and meta-analysis. *Chin Med J (Engl)*. 2023;136:34–44.

**Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

